ENN581 Robot Motion, Control and Planning · Mock exam 1 — solutions

Model answers with an indicative marking scheme. Every number was verified by script (site/src/checks/mock.py).

Mock exam 1 indicative total 64 marks · 2 hours

Q1 · Forward kinematics of a SCARA arm derive calculate 6 marks

A SCARA arm stands on a column of height H = 0.40 m. Joint 1 rotates about the vertical axis of the column. A horizontal link of length a_1 = 0.35 m leads to joint 2, which also rotates about a vertical axis. A second horizontal link a_2 = 0.25 m carries joint 3, a prismatic joint that extends the tool *downwards* by d_3. The tool frame {T} has z_T pointing down and x_T along link 2.

(a) Write an elementary transform sequence for ${}^0\mathbf{T}_T$ and a proximal DH table for the arm. [3]
(b) Evaluate ${}^0\mathbf{T}_T$ at $\theta_1 = 30°$, $\theta_2 = -60°$, d_3 = 0.10 m. [3]

(a) ETS: $t_z(H)\,R_z(\theta_1)\,t_x(a_1)\,R_z(\theta_2)\,t_x(a_2)\,t_z(-d_3)\,R_x(180°)$ [1 for the order, 1 for the final flip]. Proximal DH (α_{i-1}, a_{i-1}, θ_i, d_i): (0, 0, θ_1, H), (0, a_1, θ_2, 0), (180°, a_2, 0, d_3) [1]. The 180° twist makes z_3 point down, so d_3 is positive downwards.

(b) Position $x = a_1c_1 + a_2c_{12} = 0.3031 + 0.2165 = 0.5196$,
$y = a_1s_1 + a_2s_{12} = 0.175 - 0.125 = 0.050$, $z = H - d_3 = 0.30$ m [2]. Rotation
$R_z(\theta_1+\theta_2)R_x(180°)$ [1]:

$$
{}^0\mathbf{T}_T = \begin{bmatrix} 0.8660 & -0.5000 & 0 & 0.5196 \\ -0.5000 & -0.8660 & 0 & 0.0500 \\ 0 & 0 & -1 & 0.3000 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

Q2 · Inverse kinematics and workspace of the SCARA derive sketch 8 marks

For the arm of Q1: (a) derive θ_1, θ_2, d_3 for a desired tool position (x, y, z) and state how many solutions exist [4]; (b) evaluate them for (0.45, 0.20, 0.25) m [2]; (c) sketch the workspace if $0 \le d_3 \le 0.20$ m and the revolute joints are unlimited [2].

(a) $d_3 = H - z$ [1]. The rest is a planar 2R problem in the horizontal plane:
$\cos\theta_2 = \dfrac{x^2+y^2-a_1^2-a_2^2}{2a_1a_2}$, $\theta_2 = \operatorname{atan2}(\pm\sqrt{1-c_2^2}, c_2)$ [1],
$\theta_1 = \operatorname{atan2}(y,x) - \operatorname{atan2}(a_2s_2,\ a_1+a_2c_2)$ [1]. Two solutions (elbow left/right); none if $|c_2| > 1$ or z is out of the d_3 range [1].

(b) d_3 = 0.15 m; c_2 = 0.3286; (θ_1, θ_2) = (−4.689°, 70.818°) or (52.614°, −70.818°) [2].

(c) A hollow cylinder (thick-walled tube): radial range $|a_1 - a_2|$ = 0.10 m to $a_1 + a_2$ = 0.60 m, height range z = 0.20 to 0.40 m. Sketch a top view (annulus) and a side view (two rectangles), with dimensions [2].

Q3 · Mobility calculate explain 6 marks

Find the mobility of: (a) a planar five-bar linkage (five links including ground, five revolute joints) [1]; (b) a 3-RPS parallel platform (each leg: revolute at the base, prismatic, spherical at the platform) [2]; (c) a 3-SPS platform (spherical joints at both ends of each prismatic leg). Explain any correction you apply [3].

(a) $3(5 - 5 - 1) + 5 = 2$ [1].

(b) Spatial, D = 6; N = 1 + 1 + 3 × 2 = 8; J = 9; ΣF = 3(1) + 3(1) + 3(3) = 15; M = 6(8 − 9 − 1) + 15 = 3 [2].

(c) ΣF = 3(3) + 3(1) + 3(3) = 21; plain M = 6(−2) + 21 = 9 [1]. Each S–P–S leg can spin about its own axis without moving the platform: local mobility $R_M = 3$ [1]. Extended G–K: M = 9 + 0 − 3 = 6 [1]. (The platform has 6 DoF but only 3 actuators, so it is under-actuated.)

Q4 · Two cubic segments through a via point derive calculate 6 marks

A joint starts at rest at $\theta_0 = 0°$, passes through $\theta_v = 40°$ and stops at $\theta_g = 30°$. Each segment lasts 1 s and the acceleration is continuous at the via point. (a) List the conditions [2]. (b) Find all eight coefficients [3]. (c) What is the joint speed at the via point? [1]

(a) $\theta_1(0) = 0$, $\dot{\boldsymbol{\theta}}_1(0) = 0$, $\theta_1(1) = 40$, $\theta_2(0) = 40$, $\theta_2(1) = 30$, $\dot{\boldsymbol{\theta}}_2(1) = 0$, $\dot{\boldsymbol{\theta}}_1(1) = \dot{\boldsymbol{\theta}}_2(0)$, $\ddot{\boldsymbol{\theta}}_1(1) = \ddot{\boldsymbol{\theta}}_2(0)$ [2].

(b) Using the equal-duration closed form (Sample Q8): segment 1: 0, 0, 97.5, −57.5; segment 2: 40, 22.5, −75, 42.5 (degrees, seconds) [3].

(c) $a_{21} = 3(\theta_g - \theta_0)/(4t_f) = 22.5$ °/s [1]. (Peak speed in segment 1 is 55.109 °/s at t = 0.565 s.)

Q5 · Jacobian, singularity and static force derive calculate explain 7 marks

A planar 2R arm has $L_1 = 0.5$ m, $L_2 = 0.4$ m. (a) Derive its 2×2 position Jacobian and its singular configurations [3]. (b) At q = (30°, 60°), find the joint torques needed to hold a tip force F = (0, −10) N, and explain why τ_2 comes out as it does [2]. (c) Find $\dot{\mathbf{q}}$ for a tip velocity (0.1, 0) m/s at that pose [2].

(a) $\mathbf{J} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}$, $\det J = L_1 L_2 \sin q_2$. Singular at $q_2 = 0$ (stretched) and π (folded): the boundaries of the workspace [3].

(b) $\mathbf{J} = \begin{bmatrix} -0.6500 & -0.4000 \\ 0.4330 & 0 \end{bmatrix}$, $\boldsymbol{\tau} = \mathbf{J}^{\top}\mathbf{F} = (-4.330,\ 0)$ N·m, reading F as the force the tip applies to the environment [1]. If F is instead an external load acting on the tip, the holding torques are $-\mathbf{J}^{\top}\mathbf{F} = (+4.330,\ 0)$ N·m. State your reading: either earns the mark. $\tau_2 = 0$ because $q_1 + q_2 = 90°$: link 2 points straight up, so the vertical force acts along link 2 and has no moment about joint 2 [1].

(c) $\det J = 0.1732$; $\dot{\mathbf{q}} = \mathbf{J}^{-1}(0.1, 0) = (0,\ -0.25)$ rad/s [2]. Only joint 2 moves: link 2 is vertical, so rotating it moves the tip horizontally.

Q6 · Equations of motion of an RP arm derive calculate 8 marks

An RP arm moves in a vertical plane. Joint 1 rotates the arm by θ from the horizontal. Joint 2 slides a point mass m along the arm to distance d from joint 1. The arm itself is massless; gravity acts along −y. (a) Derive the equations of motion by Lagrange and identify M, C (velocity terms) and g [6]. (b) Evaluate τ_1 and f_2 for m = 2 kg at θ = 30°, d = 0.6 m, $\dot{\theta}$ = 0.5 rad/s, $\dot{d}$ = 0.2 m/s, $\ddot{\theta}$ = 1.0 rad/s², $\ddot{d}$ = −0.3 m/s² [2].

(a) Position (d cos θ, d sin θ); $v^2 = \dot{d}^2 + d^2\dot{\theta}^2$ [1]. $K = \frac{1}{2}m(\dot{d}^2 + d^2\dot{\theta}^2)$, $P = mgd\sin\theta$ [1]. Euler–Lagrange [2]:

$$\tau_1 = md^2\ddot{\theta} + 2md\dot{d}\dot{\theta} + mgd\cos\theta, \qquad f_2 = m\ddot{d} - md\dot{\theta}^2 + mg\sin\theta$$

M = diag(md², m); C = (2md$\dot{d}\dot{\theta}$ [Coriolis], −md$\dot{\theta}^2$ [centrifugal]); g = (mgd cos θ, mg sin θ) [2].

(b) τ_1 = 2(0.36)(1.0) + 2(2)(0.6)(0.2)(0.5) + 2(9.81)(0.6)cos 30° = 0.72 + 0.24 + 10.1949 = 11.1549 N·m; f_2 = 2(−0.3) − 2(0.6)(0.25) + 2(9.81)(0.5) = −0.6 − 0.3 + 9.81 = 8.9100 N [2].

Q7 · Partitioned control of one joint derive calculate 7 marks

A joint obeys $m\ddot{x} + b\dot{x} + kx = f$ with m = 2 kg, b = 0.5 N·s/m, k = 4 N/m. (a) Write a partitioned control law for trajectory following and the closed-loop error equation [3]. (b) Choose k_p, k_v for critical damping with ω_n = 8 rad/s [1]. (c) Find the steady-state error under a constant 5 N disturbance, and say how to remove it [2]. (d) Compute f at x = 0.10 m, $\dot{x}$ = −0.20 m/s when x_d = 0.15 m, $\dot{x}_d = \ddot{x}_d = 0$ [1].

(a) f = αf′ + β with α = m, β = b$\dot{x}$ + kx; f′ = $\ddot{x}_d + k_v\dot{E} + k_pE$, E = x_d − x; closed loop $\ddot{E} + k_v\dot{E} + k_pE = 0$ [3].

(b) $k_p = \omega_n^2$ = 64 s^{-2}, $k_v = 2\omega_n$ = 16 s^{-1} [1].

(c) $|E_{ss}| = f_{dist}/(k_p m)$ = 5/(64 × 2) = 0.0391 m. The sign depends on which way the disturbance pushes: the joint sags in the disturbance's direction. Add integral action (PID), keeping $k_i < k_v k_p$ = 1024 for stability [2].

(d) E = 0.05, $\dot{E}$ = 0.20: f′ = 16(0.2) + 64(0.05) = 6.4 m/s²; f = 2(6.4) + 0.5(−0.2) + 4(0.1) = 13.1 N [1].

Q8 · Differential-drive robot derive calculate explain 6 marks

Wheels of radius r = 0.08 m sit at (0, ±a) with a = 0.20 m (left wheel 1 at +a). (a) From the generalised wheel model, derive the chassis velocity from the wheel speeds [2]. (b) For $\dot{\varphi}$ = (5, 7.5) rad/s find u, $\dot{\theta}$ and the ICR [2]. (c) Find the wheel speeds for u = 0.3 m/s, $\dot{\theta}$ = −0.5 rad/s [1]. (d) Write the robot's non-holonomic constraint in the inertial frame [1].

(a) Rows $r\dot{\varphi}_1 = u - a\dot{\theta}$, $r\dot{\varphi}_2 = u + a\dot{\theta}$; no-slide gives v = 0; so $u = r(\dot{\varphi}_1 + \dot{\varphi}_2)/2$, $\dot{\theta} = r(\dot{\varphi}_2 - \dot{\varphi}_1)/(2a)$ [2].

(b) u = 0.5 m/s, $\dot{\theta}$ = 0.5 rad/s, ICR at (0, u/$\dot{\theta}$) = (0, 1.0) m, i.e. 1 m to the left [2].

(c) $\dot{\varphi}_1 = (0.3 + 0.1)/0.08 = 5.0$, $\dot{\varphi}_2 = (0.3 - 0.1)/0.08 = 2.5$ rad/s [1].

(d) $-\dot{x}\sin\theta + \dot{y}\cos\theta = 0$: no sideways velocity, not integrable to a position constraint [1].

Q9 · A* by hand `calculate` `explain` `6 marks`

Undirected edges with costs: S–A 2, S–B 5, A–C 4, A–D 7, B–D 3, C–G 6, D–G 2. Heuristic h: S 7, A 6, B 4, C 5, D 2, G 0. (a) Run A* from S to G; give the expansion order with g, h, f, and the path and cost [4]. (b) Is h admissible? Consistent? [2]

(a) Expand S (g 0, f 7) → open A (2+6 = 8), B (5+4 = 9). Expand A (f 8) → C (6+5 = 11), D (9+2 = 11). Expand B (f 9) → D improves to g 8, f 10 (parent B). Expand D (f 10) → G (10+0 = 10). Expand G (f 10): stop [3]. Path S–B–D–G, cost 10 [1].

(b) True costs to G: S 10, A 9, B 5, C 6, D 2; every h ≤ true, so admissible. Every edge satisfies h(n) ≤ c(n, m) + h(m), so consistent [2].

Q10 · Wheeled-robot dynamics (short) `explain` `4 marks`

The chassis frame {C} moves with velocity (u, v) and turns at $\dot{\theta}$. Show that the acceleration of the chassis origin, written in {C}, is $(\dot{u} - v\dot{\theta},\ \dot{v} + u\dot{\theta})$, and explain why a robot driving in a circle at constant u needs a sideways force even though v = 0.

Velocity in {O}: $\mathbf{v} = \boldsymbol{R}(\theta)[u, v]^\top$. Differentiate: $\dot{\mathbf{v}} = \boldsymbol{R}(\theta)[\dot{u}, \dot{v}]^\top + \dot{\boldsymbol{R}}(\theta)[u, v]^\top$ with $\dot{\boldsymbol{R}} = \boldsymbol{R}\,[\![0, -\dot{\theta}], [\dot{\theta}, 0]\!]$. In {C}: $(\dot{u} - v\dot{\theta},\ \dot{v} + u\dot{\theta})$ [2]. With v = 0 and constant u, the sideways acceleration is $u\dot{\theta} = u^2/R$: the centripetal acceleration of circular motion. The wheels' no-slip friction must supply that force [2]. (This is the term printed as $v\dot{\theta}$ on W11 slides 24–25; see the W11 tab.)